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Inventor(s)	Reiter; Tomas et al.

POWER MODULE

Abstract

A power module includes: a substrate having a patterned metallization on an electrically insulative body; first power transistor dies of a first transistor type attached to the substrate; and second power transistor dies of a second transistor type different than the first transistor type attached to the substrate. A first one or more of the first power transistor dies and a first one or more of the second power transistor dies are electrically connected in parallel via the patterned metallization to form a first topological switch. A second one or more of the first power transistor dies and a second one or more of the second power transistor dies are electrically connected in parallel via the patterned metallization to form a second topological switch. The first topological switch and the second topological switch are electrically connected in series via the patterned metallization. Additional power module embodiments are described.

Inventors: **Reiter; Tomas (Ottobrunn, DE), Niendorf; Michael (Verl, DE), Bräutigam; Philipp (Warstein-Hirschberg, DE), Jakobi; Waldemar (Soest, DE)**

Applicant: **Infineon Technologies AG** (Neubiberg, DE)

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Background/Summary

BACKGROUND

[0001] Power semiconductor modules include power semiconductor dies attached to one or more substrates and electrically interconnected within the module to form a power converter component of a power electronics device, e.g., such as a half bridge, full bridge, or B6-bridge. Conventional power semiconductor modules typically have a negative gate feedback loop that slows down switching events due to stray inductance. During di/dt switching, the negative gate feedback loop reduces the effective gate voltage and current provided by the gate driver, therefore reducing the switching speed.

[0002] Hence, there is a need for a power module design that has improved di/dt switching behavior.

SUMMARY

[0003] According to an embodiment of a power module, the power module comprises: a substrate comprising a patterned metallization on an electrically insulative body; a plurality of first power transistor dies of a first transistor type attached to the substrate; and a plurality of second power transistor dies of a second transistor type different than the first transistor type attached to the substrate, wherein a first one or more of the first power transistor dies and a first one or more of the second power transistor dies are electrically connected in parallel via the patterned metallization to form a first topological switch, wherein a second one or more of the first power transistor dies and a second one or more of the second power transistor dies are electrically connected in parallel via the patterned metallization to form a second topological switch, wherein the first topological switch and the second topological switch are electrically connected in series via the patterned metallization.

[0004] According to an embodiment of a multi-phase power electronics assembly, the multi-phase power electronics assembly comprises a plurality of power modules. The power modules can be attached to a common baseplate or coolant system. Each of the power modules supports a different phase and comprises: a substrate comprising a patterned metallization on an electrically insulative body; a plurality of first power transistor dies of a first transistor type attached to the substrate; and a plurality of second power transistor dies of a second transistor type different than the first transistor type attached to the substrate, wherein a first one or more of the first power transistor dies and a first one or more of the second power transistor dies are electrically connected in parallel via the patterned metallization to form a first topological switch, wherein a second one or more of the first power transistor dies and a second one or more of the second power transistor dies are electrically connected in parallel via the patterned metallization to form a second topological switch, wherein the first topological switch and the second topological switch are electrically connected in series via the patterned metallization and enable a load current path for the phase supported by the power module.

[0005] According to another embodiment of a power module, the power module comprises: a substrate comprising a patterned metallization on an electrically insulative body; a first power transistor die having a drain/collector pad attached to a first metallic island of the patterned metallization; and a second power transistor die having a drain/collector pad attached to the first metallic island of the patterned metallization, wherein a second metallic island of the patterned metallization is electrically connected to a source/emitter pad at a side of both the first power transistor die and the second power transistor die that faces away from the substrate, wherein a

third metallic island of the patterned metallization is electrically connected to the source/emitter pad of the first power transistor die and of the second power transistor die, wherein a load current path of the power module includes the first and second metallic islands but excludes the third metallic island.
[0006] Those skilled in the art will recognize additional features and advantages upon reading the following detailed description, and upon viewing the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE FIGURES
[0007] The elements of the drawings are not necessarily to scale relative to each other. Like reference numerals designate corresponding similar parts. The features of the various illustrated embodiments can be combined unless they exclude each other. Embodiments are depicted in the drawings and are detailed in the description which follows.
[0008] FIG. **1** illustrates a schematic view of an embodiment of a power electronics assembly that includes a power module and a gate driver IC.
[0009] FIG. **2** illustrates a top plan view of the power module, with an outline of the module frame so that the internal components of the module are visible.
[0010] FIG. **3** illustrates the thermal behavior of the power module in a motor acceleration mode.
[0011] FIG. **4** illustrates the thermal behavior of the power module in a diode (freewheeling) mode.
[0012] FIGS. **5** through **9** each illustrate a top plan view of the power module, according to additional embodiments.
[0013] FIG. **10** illustrates a top plan view of a multi-phase power electronics assembly that includes a plurality of the power modules.
DETAILED DESCRIPTION
[0014] The embodiments described herein provide a power module design without a negative gate feedback loop, and therefore has improved di/dt switching behavior. The negative gate feedback loop is omitted by providing gate driver connections that are outside the load current path of the power module. Since no load current traverses the gate driver connections, the gate driver connections to the power module are free from di/dt during switching and therefore do not experience a voltage drop. This means that the gate driver can quickly switch the power transistor dies included in the power module without negative feedback. Separately or in combination, different types of power transistor dies may be included in the power module to leverage the benefits associated with different technology types.
[0015] Described next, with reference to the figures, are exemplary embodiments of the power module design and corresponding methods of production. Any of the power module embodiments described herein may be used interchangeably unless otherwise expressly stated.
[0016] FIG. **1** illustrates a schematic view of an embodiment of a power electronics assembly that includes a power module **100** and a gate driver IC (integrated circuit) **102**. Other components of the power electronics assembly are not shown, to emphasize the power module **100**. The power electronics assembly may be used in various power applications such as a DC/AC inverter, a DC/DC converter, an AC/DC converter, a DC/AC converter, an AC/AC converter, a multi-phase inverter, an H-bridge, DC motor drive, etc.
[0017] In FIG. **1**, the power module **100** includes a first topological switch **104** and a second topological switch **106** electrically connected in series, e.g., in a half bridge configuration. Each topological switch **104**, **106** includes two or more power transistor dies (chips) connected in parallel and which are controlled simultaneously to implement the function of a single switch device. Each power transistor die in FIG. **1** is represented by a transistor symbol HS**1**, HS**2**, LS**1**, LS**2**. The first topological switch **104** may be implemented by power transistor dies HS**1**, HS**2** of

different transistor types, and the second topological switch **106** may be implemented by power transistor dies LS**1**, LS**2** of different transistor types. Different transistor types means the power transistor dies HS**1**/HS**2**, LS**1**/LS**2** that form the same topological switch **104**, **106** may utilize different semiconductor technologies (e.g., Si and SiC) and/or different transistor technologies (e.g., MOSFET and IGBT).

[0018] For example, SiC MOSFET dies and Si IGBT dies may be used to form the first topological switch **104** and/or the second topological switch **106**, with a diode die electrically connected antiparallel with each Si IGBT die. In another example, SiC MOSFET dies and Si reverse-conducting IGBT dies may be used to form the first topological switch **104** and/or the second topological switch **106**. In yet another example, Si MOSFET dies and Si IGBT dies may be used to form the first topological switch **104** and/or the second topological switch **106**. In still another example, Si IGBT dies and Si reverse-conducting IGBT dies may be used to form the first topological switch **104** and/or the second topological switch **106**. Still other types of semiconductor technologies (e.g., GaN) and/or transistor technologies (e.g., JFET—junction field-effect transistor) may be used to form the first topological switch **104** and/or the second topological switch **106**.

[0019] Using IGBT (insulated gate bipolar transistor) power semiconductor dies offers high reliability, high production yield and a favorable cost-to-current ratio. Using SiC MOSFET (metal-oxide-semiconductor field-effect transistor) power semiconductor dies offers higher efficiency and lower battery cost compared to their IGBT counterparts, since SiC MOSFET power semiconductor dies have no pn junction voltage drop and therefore low conduction losses at light load conditions. Accordingly, power modules that utilize SiC MOSFET power semiconductor dies have a more favorable WLTP (Worldwide harmonized Light vehicles Test Procedure) drive cycle efficiency compared to power modules that utilize IGBT power semiconductor dies. However, SiC MOSFET power semiconductor dies cost significantly more than IGBT power semiconductor dies. Accordingly, the power module **100** may utilize a combination of SiC MOSFET power semiconductor dies and IGBT power semiconductor dies to implement the topological switches **104**, **106** to realize higher efficiency increase in WLTP drive cycle while reducing SiC chip area and therefore decreasing chip/module cost.

[0020] In FIG. **1**, the drain/collector of each power transistor die HS**1**, HS**2** used to form the first topological switch **104** forms a positive load terminal P**1** of the power module **100**, e.g., such as a DC+ terminal in the case of a motor drive application. The source/emitter of each power transistor die LS**1**, LS**2** used to form the second topological switch **106** forms a negative load terminal N**1** of the power module **100**, e.g., such as a DC-terminal in the case of a motor drive application. The source/emitter of each power transistor die HS**1**, HS**2** used to form the first topological switch **104** and the drain/collector of each power transistor die LS**1**, LS**2** used to form the second topological switch **106** are connected to form a switch node SW of the power module **100**, e.g., such as an AC/phase terminal in the case of a motor drive application.

[0021] Each power transistor die HS**1**, HS**2**, LS**1**, LS**2** also has a gate terminal Gn.m and an auxiliary source/emitter terminal E/Sm which are used by the gate driver **102** to provide a transistor drive voltage signal that switches the corresponding topological switch **104**, **106**. At least one of the power transistor dies HS**1**, HS**2** used to form the first topological switch **104** and at least one of the power transistor dies LS**1**, LS**2** used to form the second topological switch **106** have a sense terminal TSn.m for measuring current, e.g., via a current sense element Rsense and/or temperature, e.g., via a temperature sense element Tsense at the gate driver **102**.

[0022] The gate driver **102** also provides a reference ground GND**2** for the power module **100** and may have additional outputs and inputs which are not shown in FIG. **1**, for ease of illustration. FIG. **1** shows the gate driver **102** connected to just the second topological switch **106** included in the power module **100**, for ease of illustration. The same or different gate drivers **102** may be used to switch the topological switches **104**, **106** included in the power module **100**.

[0023] FIG. **2** illustrates a top plan view of the power module **100**, with an outline of the module frame **200** so that the internal components of the module **100** are visible. The power module **100** includes a substrate **202** having a patterned first metallization **204** on an electrically insulative body **206**. The substrate **202** may also have a second metallization **208** on the opposite side of the electrically insulative body **206** as the patterned first metallization **204**. The substrate **202** may be a direct copper bonded (DCB) substrate, an active metal brazed (AMB) substrate, or an insulated metal (IMS) substrate, where in each case the electrically insulative body **206**, e.g., a ceramic body, separates the first and second metallizations **204**, **208** of the substrate **202** from one another.

[0024] The first metallization **204** of the substrate **202** is patterned to ensure proper isolation and signal routing for implementing a power electronics device implemented using the power module **100**. Exemplary electrical connections are described in more detail later in the context of a half bridge. However, a half bridge is just one example of a power electronics device that may be implemented using the power module **100**. The first metallization **204** of the substrate **202** may be patterned differently than what is illustrated in the figures, to facilitate electrical connections for any type of power electronics device implemented using the power module **100**.

[0025] The power module **100** also includes first power transistor dies **210** of a first transistor type attached to the substrate **202** and second power transistor dies **212** of a second transistor type different than the first transistor type attached to the substrate **202**. That is, the first power transistor dies **210** and the second power transistor dies **212** utilize different semiconductor technologies (e.g., Si and SiC) and/or different transistor technologies (e.g., MOSFET and IGBT).

[0026] For example, the first power transistor dies **210** may be SiC MOSFET dies and the second power transistor dies **212** may be Si IGBT dies, with a diode die **214** electrically connected antiparallel with each Si IGBT die **212** to provide a freewheeling current path. In another example, the first power transistor dies **210** may be SiC MOSFET dies and the second power transistor dies **212** may be Si reverse-conducting IGBT dies. In yet another example, the first power transistor dies **210** may be Si MOSFET dies and the second power transistor dies **212** may be Si IGBT dies. In still another example, the first power transistor dies **210** may be Si IGBT dies and the second power transistor dies **212** may be Si reverse-conducting IGBT dies. Still other types of semiconductor technologies (e.g., GaN) and/or transistor technologies (e.g., JFET-junction field-effect transistor) may be used.

[0027] In one embodiment, the power transistor dies **210**, **212** are vertical power transistor dies. For a vertical power transistor die, the primary current flow path is between the front and back sides of each die **210**, **212** (along the z direction in FIG. **2**). The drain pad is typically disposed at the die backside, with gate and source pads (and optionally one or more sense pads) at the die frontside. Additional types of semiconductor dies may be included in the power module **100**, such as logic dies, controller dies, gate driver dies, etc.

[0028] A first one or more of the first power transistor dies **210_1** and a first one or more of the second power transistor dies **212_1** are electrically connected in parallel via the patterned substrate metallization **204** to form the second topological switch **106**. FIG. **2** shows four (4) first power transistor dies **210_1** and one (1) second power transistor die **212_1** forming the second topological switch **106**. This is just an example, however. The number of first power transistor dies **210_1** and the number of second power transistor dies **212_1** that form the second topological switch **106** depend on various considerations such as the transistor types, the load requirements, etc.

[0029] A second one or more of the first power transistor dies **210_2** and a second one or more of the second power transistor dies **212_2** are electrically connected in parallel via the patterned substrate metallization **204** to form the first topological switch **104**. FIG. **2** shows four (4) first power transistor dies **210_2** and one (1) second power transistor die **212_2** forming the first topological switch **104**. This is just an example, however. The number of first power transistor dies **210_2** and the number of second power transistor dies **212_2** that form the first topological switch **104** depend on various considerations such as the transistor types, the load requirements, etc.

[0030] The first topological switch **104** and the second topological switch **106** are electrically connected in series via the patterned substrate metallization **204**, e.g., in a half bridge configuration. In the case of a half bridge configuration, the first topological switch **104** may form the high-side switch device of the half bridge and the second topological switch **106** may form the low-side switch device of the half bridge.

[0031] Continuing with the half bridge example, the patterned substrate metallization **204** may include a first metallic island **216** that provides a phase (AC) terminal for providing a phase or quasi-AC current path to the switch node SW between the first topological switch **104** and the second topological switch **106**. The drain/collector pad (out of view) of the first one or more of the first power transistor dies **210_1** and the drain/collector pad (out of view) of the first one or more of the second power transistor dies **212_1** are attached to the first metallic island **216** of the patterned substrate metallization **204**. The configuration of the patterned substrate metallization **204** may be designed for other types of power circuit configurations. For example, the power module **100** may implement a single topological switch by omitting the second one or more of the first power transistor dies **210_2** and the second one or more of the second power transistor dies **212_2**.

[0032] The patterned substrate metallization **204** may implement a high-side (DC+) power terminal of the half bridge by way of second and third metallic islands **218**, **220**. The drain/collector pad (out of view) of the second one or more of the first power transistor dies **210_2** is attached to the second metallic island **218** of the patterned substrate metallization **204**. The drain/collector pad (out of view) of the second one or more of the second power transistor dies **212_2** is attached to the third metallic island **220** of the patterned substrate metallization **204**. According to this embodiment, the high-side (DC+) power terminal of the half bridge has a split configuration. Alternatively, the patterned substrate metallization **204** may implement the high-side (DC+) power terminal of the half bridge via a single metallic island. However, in FIG. **2**, the split DC+ terminal configuration has the benefit of reduced stray inductance.

[0033] More particularly regarding the reduced stray inductance, a fourth metallic island **222** of the patterned substrate metallization **204** forms the low-side (DC−) power terminal of the half bridge. According to this embodiment, the fourth metallic island **222** of the patterned substrate metallization **204** is electrically connected to the source/emitter pad **224** of the first one or more of the first power transistor dies **210_1** and to the source/emitter pad **226** of the first one or more of the second power transistor dies **212_1** by respective electrical conductors **228**, **230** such as wire bonds, wire ribbons, metal clips, etc. In FIG. **2**, the fourth metallic island **222** of the patterned substrate metallization **204** is interposed between the second and third metallic islands **218**, **220** of the patterned substrate metallization **204** along a first end of the substrate **200** in a first lateral direction (x direction in FIG. **2**). Since the fourth metallic island **222** is at DC-potential and the second and third metallic islands **218**, **220** are at DC+ potential in this example, the stray inductance of the commutation (load current) path of the power module **100** is reduced.

[0034] In FIG. **2**, the first metallic island **216** of the patterned substrate metallization **204** runs between the second and third metallic islands **218**, **220** of the patterned substrate metallization **204** in a direction (y direction in FIG. **2**) heading toward a second end of the substrate **200** opposite the first end. The first metallic island **216** of the patterned substrate metallization **204** terminates at the second end of the substrate **200** to form the AC (phase) terminal of the power module **100** in this example.

[0035] In FIG. **2**, the patterned substrate metallization **204** also includes a fifth metallic island **232** interposed between a sixth metallic island **234** and a seventh metallic island **236** in the first lateral direction (x direction in FIG. **2**). The fifth metallic island **232** is electrically connected to the source/emitter pad **224** of the first one or more of the first power transistor dies **210_1** and to the source/emitter pad **226** of the first one or more of the second power transistor dies **212_1** by respective electrical conductors **238** such as wire bonds, wire ribbons, metal clips, etc. The sixth metallic island **234** is electrically connected to a gate pad **240** of the first one or more of the first power transistor dies **210_1** by respective electrical conductors **242** such as wire bonds, wire ribbons, metal clips, etc. The seventh metallic island **236** is electrically connected to a gate pad **244** of the first one or more of the second power transistor dies **212_1** by respective electrical conductors **246** such as wire bonds, wire ribbons, metal clips, etc.

[0036] The fifth and sixth metallic islands **232**, **234** of the patterned substrate metallization **204** enable the gate driver connection to the first one or more of the first power transistor dies **210_1**. The fifth and seventh metallic islands **232**, **236** of the patterned substrate metallization **204** enable the gate driver connection to the first one or more of the second power transistor dies **212_1**. The gate signal for the first one or more of the first power transistor dies **210_1** is carried by the sixth metallic island **234** of the patterned substrate metallization **204** and referenced to ground by the fifth metallic island **232** which is at the reference ground GND**2** provided by the gate driver **102**. The gate signal for the first one or more of the second power transistor dies **212_1** is carried by the seventh metallic island **236** of the patterned substrate metallization **204** and also referenced to ground by the fifth metallic island **232**. The fifth metallic island **232** of the patterned substrate metallization **204** forms an auxiliary source/emitter connection to the power transistor dies **210_1**, **212_1** that form the second topological switch **106**.

[0037] The auxiliary source/emitter connection implemented by the fifth metallic island **232** of the patterned substrate metallization **204** ensures there is no negative gate feedback loop for the second topological switch **106**. The fourth metallic island **222** of the patterned substrate metallization **204** has the same steady state potential as the fifth metallic island **232**. However, the fourth metallic island **222** carries a switched load current during switching of the second topological switch **106** and therefore has a voltage drop over its trace/length given by dV=L*di/dt, where L is the stray inductance. In contrast, no load current flows in any of the fifth, sixth and seventh metallic islands **232**, **234**, **236**. That is, the load current path of the power module **100**, which is illustrated in FIG. **2** by dashed arrows, includes the first through fourth metallic islands **216**, **218**, **220**, **222** but excludes the fifth, sixth and seventh metallic islands **232**, **234**, **236**. Accordingly, di/dt current does occur in the fifth metallic island **232** during switching and therefore no voltage drop across along the trace/length of the fifth metallic island **232**. This means no gate voltage or current drop and the gate driver **102** can switch the power transistor dies **210_1**, **212_1** that form the second topological switch **106** at a high frequency without negative feedback.

[0038] If the first topological switch **104** is also included in the power module **100**, the patterned substrate metallization **204** can be configured to avoid a negative gate feedback loop for the first topological switch **104**. For example, in FIG. **2**, an eighth metallic island **248** of the patterned substrate metallization **204** is interposed between a ninth metallic island **250** and a tenth metallic island **252** of the patterned substrate metallization **204** in the first lateral direction (x direction in FIG. **2**). The eighth metallic island **248** is electrically connected to a source/emitter pad **254** of the second one or more of the first power transistor dies **210_2** and to a source/emitter pad **256** of the second one or more of the second power transistor dies **212_2** by one or more respective electrical conductors **255**, **257** such as wire bonds, wire ribbons, metal clips, etc. The source/emitter pad **254** of the second one or more of the first power transistor dies **210_2** and the source/emitter pad **256** of the second one or more of the second power transistor dies **212_2** are electrically connected to the first metallic island **216** of the patterned substrate metallization **204** by respective electrical conductors **258**, **260** such as wire bonds, wire ribbons, metal clips, etc.

[0039] The ninth metallic island **250** of the patterned substrate metallization **204** is electrically connected to a gate pad **262** of the second one or more of the first power transistor dies **210_2** by one or more electrical conductors **264** such as wire bonds, wire ribbons, metal clips, etc. The tenth metallic island **252** of the patterned substrate metallization **204** is electrically connected to a gate pad **266** of the second one or more of the second power transistor dies **212_2** by one or more electrical conductors **268** such as wire bonds, wire ribbons, metal clips, etc.

[0040] The eighth and ninth metallic islands **248**, **250** of the patterned substrate metallization **204** enable the gate driver connection to the second one or more of the first power transistor dies **210_2**. The eighth and tenth metallic islands **248**, **252** of the patterned substrate metallization **204** enable the gate driver connection to the second one or more of the second power transistor dies **212_2**. The gate signal for the second one or more of the first power transistor dies **210_2** is carried by the ninth metallic island **250** of the patterned substrate metallization **204** and referenced to ground by the eighth metallic island **248** which is at the reference ground GND**2** provided by the gate driver **102**. The gate signal for the second one or more of the second power transistor dies **212_2** is carried by the tenth metallic island **252** of the patterned substrate metallization **204** and also referenced to ground by the eighth metallic island **248**. The eighth metallic island **248** of the patterned substrate metallization **204** forms an auxiliary source/emitter connection to the power transistor dies **210_2**, **212_2** that form the first topological switch **104**.

[0041] The auxiliary source/emitter connection implemented by the eighth metallic island **248** of the patterned substrate metallization **204** ensures there is no negative gate feedback loop for the first topological switch **104**. The first metallic island **216** of the patterned substrate metallization **204** has the same steady state potential as the eighth metallic island **248**. However, the first metallic island **216** carries a switched load current during switching of the first topological switch **104** and therefore has a voltage drop over its trace/length given by dV=L*di/dt, where L is the stray inductance. In contrast, no load current flows in the eighth, ninth and tenth metallic islands **248**, **250**, **252**. That is, the load current path of the power module **100** includes the first through fourth metallic islands **216**, **218**, **220**, **222** but excludes the eighth, ninth and tenth metallic islands **248**, **250**, **252**. Accordingly, di/dt current does not occur in the eighth metallic island **248** during switching and therefore no voltage drop across along the trace/length of the eighth metallic island **248**. This means no gate voltage or current drop and the gate driver **102** can switch the power transistor dies **210_2**, **212_2** that form the first topological switch **104** at a high frequency without negative feedback. As explained above, the power module **100** instead may include only one of the topological switches **104**, **106**.

[0042] In FIG. **2**, the eighth, ninth and tenth metallic islands **248**, **250**, **252** of the patterned substrate metallization **204** are interposed between a first branch **216_1** and a second branch **216_2** of the first metallic island **216** of the patterned substrate metallization **204** in the first lateral direction (x direction in FIG. **2**). The first branch **216_1** of the first metallic island **216** is electrically connected to the source/emitter pad **254** of the second one or more of the first power transistor dies **210_2** by one or more first electrical conductors **258** such as wire bonds, wire ribbons, metal clips, etc. The second branch **216_2** of the first metallic island **216** is electrically connected to the source/emitter pad **256** of the second one or more of the second power transistor dies **212_2** by one or more second electrical conductors **260** such as wire bonds, wire ribbons, metal clips, etc.

[0043] Likewise, the fifth, sixth and seventh metallic islands **232**, **234**, **236** of the patterned substrate metallization **204** are interposed between a first branch **222_1** and a second branch **222_2** of the fourth metallic island **222** of the patterned substrate metallization **204** in the first lateral direction (x direction in FIG. **2**). The first branch **222_1** of the fourth metallic island **222** is electrically connected to the source/emitter pad **224** of the first one or more of the first power transistor dies **210_1** by one or more first electrical conductors **228** such as wire bonds, wire ribbons, metal clips, etc. The second branch **222_2** of the fourth metallic island **222** is electrically connected to the source/emitter pad **226** of the first one or more of the second power transistor dies **212_1** by one or more second electrical conductors **230** such as wire bonds, wire ribbons, metal clips, etc.

[0044] In FIG. **2**, the fourth metallic island **222** of the patterned substrate metallization **204** is interposed in the first lateral direction (x direction in FIG. **2**) between a metallic island **270** of the patterned substrate metallization **204** that is electrically connected to a sense pad **272** of a single one of the first one or more of the first power transistor dies **210_1** by one or more electrical conductors **274** such as wire bonds, wire ribbons, metal clips, etc. and the metallic island **234** of the patterned substrate metallization **204** that is electrically connected to the gate pad **240** of the first one or more of the first power transistor dies **210_1**. The fourth metallic island **222** of the patterned substrate metallization **204** also is interposed in the first lateral direction (x direction in FIG. **2**) between a metallic island **276** of the patterned substrate metallization **204** that is electrically connected to a sense pad **278** of a single one of the first one or more of the second power transistor dies **212_1** by one or more electrical conductors **280** such as wire bonds, wire ribbons, metal clips, etc. and the metallic island **236** of the patterned substrate metallization **204** that is electrically connected to the gate pad **244** of the first one or more of the second power transistor dies **212_1**.

[0045] In a similar manner for the second topological switch **106**, if included in the power module **100**, the first metallic island **216** of the patterned substrate metallization **204** may be interposed in the first lateral direction (x direction in FIG. **2**) between a metallic island **282** of the patterned substrate metallization **204** that is electrically connected to a sense pad **284** of a single one of the second one or more of the first power transistor dies **210_2** by one or more electrical conductors **286** such as wire bonds, wire ribbons, metal clips, etc. and the metallic island **250** of the patterned substrate metallization **204** that is electrically connected to the gate pad **262** of the second one or more of the first power transistor dies **210_2**. The first metallic island **216** of the patterned substrate metallization **204** also may be interposed in the first lateral direction (x direction in FIG. **2**) between a metallic island **288** of the patterned substrate metallization **204** electrically that is electrically connected to a sense pad **290** of a single one of the second one or more of the second power transistor dies **212_2** by one or more electrical conductors **292** such as wire bonds, wire ribbons, metal clips, etc. and the metallic island **252** of the patterned substrate metallization **204** that is electrically connected to the gate pad **266** of the second one or more of the second power transistor dies **212_2**.

[0046] In FIG. **2**, a single sensor pin **294** is attached to the metallic island **270** of the patterned substrate metallization **204** that is electrically connected to the sense pad **272** of the single one of the first one or more of the first power transistor dies **210_1** and a single sensor pin **296** is attached to the metallic island **276** of the patterned substrate metallization **204** that is electrically connected to the sense pad **278** of the single one of the first one or more of the second power transistor dies **212_1**. A single sensor pin **298** is attached to the metallic island **282** of the patterned substrate metallization **204** that is electrically connected to the sense pad **284** of the single one of the second one or more of the first power transistor dies **210_2** and a single sensor pin **300** is attached to the metallic island **288** of the patterned substrate metallization **204** that is electrically connected to the sense pad **290** of the single one of the second one or more of the second power transistor dies **212_2**. Additional pins **302** may be attached to the patterned substrate metallization **204** to facilitate further external electrical connections to the components housed in the power module **100**, e.g., gate connections, auxiliary source/emitter connections, etc.

[0047] Alternatively, only one of the first power transistor dies **210_1**, **212_1** that forms the second topological switch **106** and/or only one of the power transistor dies **210_2**, **212_2** that forms the first topological switch **104** may have a sense pad **272**/**278**, so that one of the sense pads **272**, **278** for the second topological switch **106** and/or one of the sense pads **284**, **290** for the first topological switch **104** (and the corresponding connection and pin) shown in FIG. **2** may be omitted. That is, only one of the transistor types may provide the sense functionality, e.g., IGBTs and not SiC MOSFETs or vice-versa. All of the power transistor dies **210**, **212** may have a sense pad but only one sense pad from each group of power transistor dies **210_1**, **212_1**, **210_2**, **212_2**, may be connected to the patterned substrate metallization **204**.

[0048] More generally, any of the power transistor dies **210**, **212** that form either of the topological switches **104**, **106** may implement sensor functionality such as a temperature sensor, a current mirror, or a combined temperature sensor and current mirror. The corresponding module sense pin **294**, **296**, **298**, **300** may be located close to the corresponding gate connection of the corresponding switch but so as to not disturb the load current path for minimized stray inductances and resistances, the load current path may be between the corresponding gate and sensor pin locations. As explained above, only one power transistor die **210**, **212** per topological switch **104**, **106** may implement the sense functionality to save pin connection and layout/routing effort. However, more than power transistor die **210**, **212** per topological switch **104**, **106** may implement sense functionality.

[0049] In FIG. **2**, the first one or more of the first power transistor dies **210_1** and the second one or more of the first power transistor dies **210_2** are arranged diagonally with respect to one another on the module substrate **202**, and the first one or more of the second power transistor dies **212_1** and the second one or more of the second power transistor dies **212_2** are also arranged diagonally with respect to one another on the module substrate **202**. That is, the high-side and low-side power transistor dies **210**, **212** of the same transistor type are offset/arranged diagonal to one another to provide more separation distance without having to increase the size of the module substrate **202**. Such a configuration is optimal for thermal coupling. For example, in the case of a motor application, when in diode (freewheeling) mode, e.g., when stopping the motor, only the diode dies are loaded. When accelerating the motor, the IGBTs **212** are mainly loaded. The diagonal arrangement shown in FIG. **2** provides the greatest distance between the IGBT dies **212** and the diode dies **214** of the high-side and low-side switches **104**, **106**. Accordingly, the hottest components are separated by the largest distances.

[0050] FIG. **3** illustrates the thermal behavior of the power module **100** in a motor acceleration mode, e.g., when accelerating a motor load powered by the module **100**. The hottest die components in the acceleration mode are indicated by the boxes labelled **400** in FIG. **3**. The next hottest die components in the acceleration mode are indicated by the boxes labelled **402** in FIG. **3**. The coolest die components in the acceleration mode are indicated by the boxes labelled **404** in FIG. **3**.

[0051] FIG. **4** illustrates the thermal behavior of the power module in a diode (freewheeling) mode, e.g., when stopping a motor load powered by the module **100**. The hottest die components in the diode mode are indicated by the boxes labelled **400** in FIG. **4**. The next hottest die components in the diode mode are indicated by the boxes labelled **402** in FIG. **4**. The coolest die components in the diode mode are indicated by the boxes labelled **404** in FIG. **4**.

[0052] FIG. **5** illustrates a top plan view of the power module **100** with the module frame **200** omitted, according to another embodiment. In FIG. **5**, the fifth metallic island **232** of the patterned substrate metallization **204** is interposed between the sixth and seventh metallic islands **234**, **236** of the patterned substrate metallization **204** in a second lateral direction (y direction in FIGS. **2** and **5**) orthogonal to the first lateral direction (x direction in FIGS. **2** and **5**). Likewise, the eighth metallic island **248** of the patterned substrate metallization **204** is interposed between the ninth and tenth metallic islands **250**, **252** of the patterned substrate metallization **204** in the second lateral direction (y direction in FIGS. **2** and **5**) in FIG. **5**.

[0053] FIG. **6** illustrates a top plan view of the power module **100** with the module frame **200** omitted, according to another embodiment. In FIG. **6**, the first metallic island **216** of the patterned substrate metallization **204** is electrically connected to an additional metallic island **500** of the patterned substrate metallization **204** by one or more electrical conductors **502** such as wire bonds, wire ribbons, metal clips, etc. The additional metallic island **500** terminates at the second end of the module substrate **202** and forms the AC/phase module terminal at the opposite side of the substrate **204** as the metallic islands **218**, **220** that form the DC+ module terminal and the metallic island **222** that forms the DC-module terminal.

[0054] FIG. **7** illustrates a top plan view of the power module **100** with the module frame **200** omitted, according to another embodiment. In FIG. **7**, the source/emitter pad **254** of the second one or more of the first power transistor dies **210_2** is electrically connected to the first metallic island **216** of the patterned substrate metallization **204** by one or more respective electrical conductors

255 such as wire bonds, wire ribbons, metal clips, etc. Likewise, the source/emitter pad **256** of the second one or more of the second power transistor dies **212_2** is electrically connected to the first metallic island **216** of the patterned substrate metallization **204** by one or more respective electrical conductors **257** such as wire bonds, wire ribbons, metal clips, etc. in FIG. **7**.

[0055] Also in FIG. **7**, the first one or more of the first power transistor dies **210_1** includes at least three of the first power transistor dies **210** arranged in a first row **600** on the same metallic island **602** of the patterned substrate metallization **204**. Likewise, the second one or more of the first power transistor dies **210_2** includes at least three of the first power transistor dies **210** arranged in a second row **604** on the same metallic island **606** of the patterned substrate metallization **204**. The first row **600** of the first power transistor dies **210** is shown attached to the first metallic island **216** of the patterned substrate metallization **204** and the second row **604** of the first power transistor dies **210** is shown attached to the second metallic island **218** of the patterned substrate metallization **204** in FIG. **7**. However, the first row **600** of the first power transistor dies **210** instead may be attached to a different metallic island of the patterned substrate metallization **204** than the first metallic island **216** and the second row **604** of the first power transistor dies **210** may be attached to a different metallic island of the patterned substrate metallization **204** than the second metallic island **218**.

[0056] FIG. **8** illustrates a top plan view of the power module **100** with the module frame **200** omitted, according to another embodiment. In FIG. **8**, the metallic island **602** of the patterned substrate metallization **204** to which the first row **600** of the first power transistor dies **210** is attached has a wider part **602_1** to which each intermediate (inner) one of the at least three of the first power transistor dies **210** is attached and a narrower part **602_2** to which end (outer) ones of the at least three of the first power transistor dies **210** are attached, to better balance thermal dissipation. That is, each intermediate one of the at least three of the first power transistor dies **210** in the first row **600** has a higher thermal capacitance than the end dies **210** in the first row **600** because of the larger surrounding metal area. Likewise, the metallic island **606** of the patterned substrate metallization **204** to which the second row **604** of the first power transistor dies **210** is attached has a wider part **606_1** to which each intermediate (inner) one of the at least three of the first power transistor dies **210** is attached and a narrower part **606_2** to which end (outer) ones of the at least three of the first power transistor dies **210** are attached.

[0057] Also in FIG. **8**, a first one or more electrical conductors **228_1** connects the source/emitter pad **224** of each intermediate one of the at least three of the first power transistor dies **210** in the first row **600** to the fourth metallic island **222** of the patterned substrate metallization **204**. A second one or more electrical conductors **228_2** connects the source/emitter pad **224** of each end one of the at least three of the first power transistor dies **210** in the first row **600** to the fourth metallic island **222**. The first one or more electrical conductors **228_1** are longer (and thus have higher resistance) than the second one or more electrical conductors **228_2**, because the metallic island **602** to which the first row **600** of the first power transistor dies **210** is attached has a wider intermediate part **602_1** and a narrower end part **602_2**. This means that current sharing among the first power transistor dies **210** in the first row **600** is slightly asymmetric. Each intermediate one of the at least three of the first power transistor dies **210** in the first row **600** intermediate has lower current and therefore dissipates less heat (i.e., has lower power losses) and thus has better cooling, avoiding a thermal hotspot at each intermediate die **210** in the first row **600**. The sensor function provided by one of the end dies **210** in the first row **600** measures a more precise temperature for the corresponding topological switch **106**.

[0058] Likewise, a first one or more electrical conductors **258_1** connects the source/emitter pad **254** of each intermediate one of the at least three of the first power transistor dies **210** in the second row **604** to the first metallic island **216** of the patterned substrate metallization **204**. A second one or more electrical conductors **258_2** connects the source/emitter pad **254** of each end one of the at least three of the first power transistor dies **210** in the second row **604** to the first metallic island

222. The first one or more electrical conductors **258_1** are longer (and thus have higher resistance) than the second one or more electrical conductors **258_2**, because the metallic island **606** to which the second row **604** of the first power transistor dies **210** is attached has a wider intermediate part **606_1** and a narrower end part **606_2**.
[0059] FIG. **9** illustrates a top plan view of the power module **100** with the module frame **200** omitted, according to another embodiment. In FIG. **9**, the source/emitter pad **254** of the second one or more of the first power transistor dies **210_2** is electrically connected to the eighth metallic island **248** of the patterned substrate metallization **204** instead of the first metallic island **216**. Likewise, the source/emitter pad **256** of the second one or more of the second power transistor dies **212_2** is electrically connected to the eighth metallic island **248** of the patterned substrate metallization **204** instead of the first metallic island **216**.
[0060] FIG. **10** illustrates a top plan view of a multi-phase power electronics assembly **700** that includes a plurality of the power modules **100**. Each of the power modules **100** supports a different phase U, V, W of the multi-phase power electronics assembly **700**. For example, the multi-phase power electronics assembly **700** may power a 3-phase motor and each of the power modules **100** energizes a different phase U, V, W of the motor. The power modules **100** may be attached to a common baseplate or coolant system **702**.
[0061] Although the present disclosure is not so limited, the following numbered examples demonstrate one or more aspects of the disclosure.
[0062] Example 1. A power module, comprising: a substrate comprising a patterned metallization on an electrically insulative body; a plurality of first power transistor dies of a first transistor type attached to the substrate; and a plurality of second power transistor dies of a second transistor type different than the first transistor type attached to the substrate, wherein a first one or more of the first power transistor dies and a first one or more of the second power transistor dies are electrically connected in parallel via the patterned metallization to form a first topological switch, wherein a second one or more of the first power transistor dies and a second one or more of the second power transistor dies are electrically connected in parallel via the patterned metallization to form a second topological switch, wherein the first topological switch and the second topological switch are electrically connected in series via the patterned metallization.
[0063] Example 2. The power module of example 1, wherein a drain/collector pad of the first one or more of the first power transistor dies and a drain/collector pad of the first one or more of the second power transistor dies are attached to a first metallic island of the patterned metallization, wherein a drain/collector pad of the second one or more of the first power transistor dies is attached to a second metallic island of the patterned metallization, wherein a drain/collector pad of the second one or more of the second power transistor dies is attached to a third metallic island of the patterned metallization, and wherein the second metallic island and the third metallic island form a first DC terminal of the power module.
[0064] Example 3. The power module of example 2, wherein a fourth metallic island of the patterned metallization forms a second DC terminal of the power module, and wherein the fourth metallic island is electrically connected to a source/emitter pad of the first one or more of the first power transistor dies and of the first one or more of the second power transistor dies.
[0065] Example 4. The power module of example 3, wherein along a first end of the substrate, the fourth metallic island is interposed between the second metallic island and the third metallic island.
[0066] Example 5. The power module of example 4, wherein the first metallic island runs between the second metallic island and the third metallic island in a direction heading toward a second end of the substrate opposite the first end.
[0067] Example 6. The power module of example 5, wherein the first metallic island terminates at the second end of the substrate and forms an AC terminal of the power module.
[0068] Example 7. The power module of example 5, wherein the first metallic island is connected to an additional metallic island of the patterned metallization by one or more electrical conductors,

and wherein the additional metallic island terminates at the second end of the substrate.
[0069] Example 8. The power module of any of examples 3 through 7, wherein a fifth metallic island of the patterned metallization is interposed between a sixth metallic island and a seventh metallic island of the patterned metallization, wherein the fifth metallic island is electrically connected to the source/emitter pad of the first one or more of the first power transistor dies and of the first one or more of the second power transistor dies, wherein the sixth metallic island is electrically connected to a gate pad of the first one or more of the first power transistor dies, and wherein the seventh metallic island is electrically connected to a gate pad of the first one or more of the second power transistor dies.
[0070] Example 9. The power module of example 8, wherein the fifth metallic island, the sixth metallic island and the seventh metallic island are interposed between a first branch and a second branch of the fourth metallic island, wherein the first branch of the fourth metallic island is connected to the source/emitter pad of the first one or more of the first power transistor dies by one or more first electrical conductors, and wherein the second branch of the fourth metallic island is connected to the source/emitter pad of the first one or more of the second power transistor dies by one or more second electrical conductors.
[0071] Example 10. The power module of example 8 or 9, wherein an eighth metallic island of the patterned metallization is interposed between a ninth metallic island and a tenth metallic island of the patterned metallization, wherein the eighth metallic island is electrically connected to a source/emitter pad of the second one or more of the first power transistor dies and of the second one or more of the second power transistor dies, wherein the ninth metallic island is electrically connected to a gate pad of the second one or more of the first power transistor dies, and wherein the tenth metallic island is electrically connected to a gate pad of the second one or more of the second power transistor dies.
[0072] Example 11. The power module of example 10, wherein the eighth metallic island, the ninth metallic island and the tenth metallic island are interposed between a first branch and a second branch of the first metallic island, wherein the first branch of the first metallic island is connected to the source/emitter pad of the second one or more of the first power transistor dies by one or more first electrical conductors, and wherein the second branch of the first metallic island is connected to the source/emitter pad of the second one or more of the second power transistor dies by one or more second electrical conductors.
[0073] Example 12. The power module of any of examples 1 through 11, wherein the first one or more of the first power transistor dies and the second one or more of the first power transistor dies are arranged diagonally with respect to one another on the substrate, and wherein the first one or more of the second power transistor dies and the second one or more of the second power transistor dies are arranged diagonally with respect to one another on the substrate.
[0074] Example 13. The power module of any of examples 1 through 12, wherein a metallic island of the patterned metallization that is electrically connected to a source/emitter pad of the first one or more of the first power transistor dies and of the first one or more of the second power transistor dies is interposed between a metallic island of the patterned metallization that is electrically connected to a sense pad of a single one of the first one or more of the first power transistor dies and a metallic island of the patterned metallization that is electrically connected to a gate pad of the first one or more of the first power transistor dies, and wherein the metallic island of the patterned metallization that is electrically connected to the source/emitter pad of the first one or more of the first power transistor dies and of the first one or more of the second power transistor dies is interposed between a metallic island of the patterned metallization that is electrically connected to a sense pad of a single one of the first one or more of the second power transistor dies and a metallic island of the patterned metallization that is electrically connected to a gate pad of the first one or more of the second power transistor dies.
[0075] Example 14. The power module of example 13, wherein a single sensor pin is attached to

the metallic island of the patterned metallization that is electrically connected to the sense pad of the single one of the first one or more of the first power transistor dies, and wherein a single sensor pin is attached to the metallic island of the patterned metallization that is electrically connected to the sense pad of the single one of the first one or more of the second power transistor dies.

[0076] Example 15. The power module of any of examples 1 through 14, wherein a metallic island of the patterned metallization that is electrically connected to a source/emitter pad of the second one or more of the first power transistor dies and of the second one or more of the second power transistor dies is interposed between a metallic island of the patterned metallization that is electrically connected to a sense pad of a single one of the second one or more of the first power transistor dies and a metallic island of the patterned metallization that is electrically connected to a gate pad of the second one or more of the first power transistor dies, and wherein the metallic island of the patterned metallization that is electrically connected to the source/emitter pad of the second one or more of the first power transistor dies and of the second one or more of the second power transistor dies is interposed between a metallic island of the patterned metallization that is electrically connected to a sense pad of a single one of the second one or more of the second power transistor dies and a metallic island of the patterned metallization that is electrically connected to a gate pad of the second one or more of the second power transistor dies.

[0077] Example 16. The power module of example 15, wherein a single sensor pin is attached to the metallic island of the patterned metallization that is electrically connected to the sense pad of the single one of the second one or more of the first power transistor dies, and wherein a single sensor pin is attached to the metallic island of the patterned metallization that is electrically connected to the sense pad of the single one of the second one or more of the second power transistor dies.

[0078] Example 17. The power module of any of examples 1 through 16, wherein the first power transistor dies are SiC MOSFET dies, wherein the second power transistor dies are Si IGBT dies, and wherein a diode die is electrically connected antiparallel with each Si IGBT die.

[0079] Example 18. The power module of any of examples 1 through 16, wherein the first power transistor dies are SiC MOSFET dies, and wherein the second power transistor dies are Si reverse-conducting IGBT dies.

[0080] Example 19. The power module of any of examples 1 through 16, wherein the first power transistor dies are Si MOSFET dies, and wherein the second power transistor dies are Si IGBT dies.

[0081] Example 20. The power module of any of examples 1 through 16, wherein the first power transistor dies are Si IGBT dies, and wherein the second power transistor dies are Si reverse-conducting IGBT dies.

[0082] Example 21. The power module of any of examples 1 through 20, wherein the first one or more of the first power transistor dies comprises at least three of the first power transistor dies arranged in a row on a same metallic island of the patterned metallization, and wherein a part of the metallic island to which each intermediate one of the at least three of the first power transistor dies is attached is wider than each part of the metallic island to which end ones of the at least three of the first power transistor dies are attached.

[0083] Example 22. The power module of any of examples 1 through 21, wherein a first one or more electrical conductors connects a source/emitter pad of each intermediate one of the at least three of the first power transistor dies to an additional metallic island of the patterned metallization, wherein a second one or more electrical conductors connects a source/emitter pad of each end one of the at least three of the first power transistor dies to the additional metallic island, and wherein the first one or more electrical conductors are longer than the second one or more electrical conductors.

[0084] Example 23. The power module of any of examples 1 through 22, wherein the second one or more of the first power transistor dies comprises at least three of the first power transistor dies arranged in a row on a same metallic island of the patterned metallization, and wherein a part of the metallic island to which each intermediate one of the at least three of the first power transistor dies is attached is wider than each part of the metallic island to which end ones of the at least three of the first power transistor dies are attached.

[0085] Example 24. The power module of any of examples 1 through 23, wherein a first one or more electrical conductors connects a source/emitter pad of each intermediate one of the at least three of the first power transistor dies to an additional metallic island of the patterned metallization, wherein a second one or more electrical conductors connects a source/emitter pad of each end one of the at least three of the first power transistor dies to the additional metallic island, and wherein the first one or more electrical conductors are longer than the second one or more electrical conductors.

[0086] Example 25. The power module of any of examples 1 through 24, wherein a drain/collector pad of the first one or more of the first power transistor dies and a drain/collector pad of the first one or more of the second power transistor dies are attached to a first metallic island of the patterned metallization, wherein a source/emitter pad of the first one or more of the first power transistor dies and a source/emitter pad of the first one or more of the second power transistor dies are electrically connected to a second metallic island of the patterned metallization, wherein a third metallic island of the patterned metallization is electrically connected to the source/emitter pad of the first one or more of the first power transistor dies and of the first one or more of the second power transistor dies, wherein a load current path of the power module includes the first and second metallic islands but excludes the third metallic island.

[0087] Example 26. A multi-phase power electronics assembly comprising a plurality of power modules, where each of the power modules supports a different phase and comprises: a substrate comprising a patterned metallization on an electrically insulative body; a plurality of first power transistor dies of a first transistor type attached to the substrate; and a plurality of second power transistor dies of a second transistor type different than the first transistor type attached to the substrate, wherein a first one or more of the first power transistor dies and a first one or more of the second power transistor dies are electrically connected in parallel via the patterned metallization to form a first topological switch, wherein a second one or more of the first power transistor dies and a second one or more of the second power transistor dies are electrically connected in parallel via the patterned metallization to form a second topological switch, wherein the first topological switch and the second topological switch are electrically connected in series via the patterned metallization and enable a load current path for the phase supported by the power module.

[0088] Example 27. A power module, comprising: a substrate comprising a patterned metallization on an electrically insulative body; a first power transistor die having a drain/collector pad attached to a first metallic island of the patterned metallization; and a second power transistor die having a drain/collector pad attached to the first metallic island of the patterned metallization, wherein a second metallic island of the patterned metallization is electrically connected to a source/emitter pad at a side of both the first power transistor die and the second power transistor die that faces away from the substrate, wherein a third metallic island of the patterned metallization is electrically connected to the source/emitter pad of the first power transistor die and of the second power transistor die, wherein a load current path of the power module includes the first and second metallic islands but excludes the third metallic island.

[0089] Terms such as “first”, “second”, and the like, are used to describe various elements, regions, sections, etc. and are also not intended to be limiting. Like terms refer to like elements throughout the description.

[0090] As used herein, the terms “having”, “containing”, “including”, “comprising” and the like are open ended terms that indicate the presence of stated elements or features, but do not preclude additional elements or features. The articles “a”, “an” and “the” are intended to include the plural as well as the singular, unless the context clearly indicates otherwise.

[0091] The expression “and/or” should be interpreted to cover all possible conjunctive and disjunctive combinations, unless expressly noted otherwise. For example, the expression “A and/or B” should be interpreted to mean only A, only B, or both A and B. The expression “at least one of” should be interpreted in the same manner as “and/or”, unless expressly noted otherwise. For example, the expression “at least one of A and B” should be interpreted to mean only A, only B, or both A and B.

[0092] It is to be understood that the features of the various embodiments described herein may be combined with each other, unless specifically noted otherwise.

[0093] Although specific embodiments have been illustrated and described herein, it will be appreciated by those of ordinary skill in the art that a variety of alternate and/or equivalent implementations may be substituted for the specific embodiments shown and described without departing from the scope of the present invention. This application is intended to cover any adaptations or variations of the specific embodiments discussed herein. Therefore, it is intended that this invention be limited only by the claims and the equivalents thereof.

Claims

1. A power module, comprising: a substrate comprising a patterned metallization on an electrically insulative body; a plurality of first power transistor dies of a first transistor type attached to the substrate; and a plurality of second power transistor dies of a second transistor type different than the first transistor type attached to the substrate, wherein a first one or more of the first power transistor dies and a first one or more of the second power transistor dies are electrically connected in parallel via the patterned metallization to form a first topological switch, wherein a second one or more of the first power transistor dies and a second one or more of the second power transistor dies are electrically connected in parallel via the patterned metallization to form a second topological switch, wherein the first topological switch and the second topological switch are electrically connected in series via the patterned metallization.

2. The power module of claim 1, wherein a drain/collector pad of the first one or more of the first power transistor dies and a drain/collector pad of the first one or more of the second power transistor dies are attached to a first metallic island of the patterned metallization, wherein a drain/collector pad of the second one or more of the first power transistor dies is attached to a second metallic island of the patterned metallization, wherein a drain/collector pad of the second one or more of the second power transistor dies is attached to a third metallic island of the patterned metallization, and wherein the second metallic island and the third metallic island form a first DC terminal of the power module.

3. The power module of claim 2, wherein a fourth metallic island of the patterned metallization forms a second DC terminal of the power module, and wherein the fourth metallic island is electrically connected to a source/emitter pad of the first one or more of the first power transistor dies and of the first one or more of the second power transistor dies.

4. The power module of claim 3, wherein along a first end of the substrate, the fourth metallic island is interposed between the second metallic island and the third metallic island.

5. The power module of claim 4, wherein the first metallic island runs between the second metallic island and the third metallic island in a direction heading toward a second end of the substrate opposite the first end.

6. The power module of claim 5, wherein the first metallic island terminates at the second end of the substrate and forms an AC terminal of the power module.

7. The power module of claim 5, wherein the first metallic island is connected to an additional metallic island of the patterned metallization by one or more electrical conductors, and wherein the additional metallic island terminates at the second end of the substrate.

8. The power module of claim 3, wherein a fifth metallic island of the patterned metallization is interposed between a sixth metallic island and a seventh metallic island of the patterned metallization, wherein the fifth metallic island is electrically connected to the source/emitter pad of the first one or more of the first power transistor dies and of the first one or more of the second power transistor dies, wherein the sixth metallic island is electrically connected to a gate pad of the first one or more of the first power transistor dies, and wherein the seventh metallic island is electrically connected to a gate pad of the first one or more of the second power transistor dies.

9. The power module of claim 8, wherein the fifth metallic island, the sixth metallic island and the seventh metallic island are interposed between a first branch and a second branch of the fourth metallic island, wherein the first branch of the fourth metallic island is connected to the source/emitter pad of the first one or more of the first power transistor dies by one or more first electrical conductors, and wherein the second branch of the fourth metallic island is connected to the source/emitter pad of the first one or more of the second power transistor dies by one or more second electrical conductors.

10. The power module of claim 8, wherein an eighth metallic island of the patterned metallization is interposed between a ninth metallic island and a tenth metallic island of the patterned metallization, wherein the eighth metallic island is electrically connected to a source/emitter pad of the second one or more of the first power transistor dies and of the second one or more of the second power transistor dies, wherein the ninth metallic island is electrically connected to a gate pad of the second one or more of the first power transistor dies, and wherein the tenth metallic island is electrically connected to a gate pad of the second one or more of the second power transistor dies.

11. The power module of claim 10, wherein the eighth metallic island, the ninth metallic island and the tenth metallic island are interposed between a first branch and a second branch of the first metallic island, wherein the first branch of the first metallic island is connected to the source/emitter pad of the second one or more of the first power transistor dies by one or more first electrical conductors, and wherein the second branch of the first metallic island is connected to the source/emitter pad of the second one or more of the second power transistor dies by one or more second electrical conductors.

12. The power module of claim 1, wherein the first one or more of the first power transistor dies and the second one or more of the first power transistor dies are arranged diagonally with respect to one another on the substrate, and wherein the first one or more of the second power transistor dies and the second one or more of the second power transistor dies are arranged diagonally with respect to one another on the substrate.

13. The power module of claim 1, wherein a metallic island of the patterned metallization that is electrically connected to a source/emitter pad of the first one or more of the first power transistor dies and of the first one or more of the second power transistor dies is interposed between a metallic island of the patterned metallization that is electrically connected to a sense pad of a single one of the first one or more of the first power transistor dies and a metallic island of the patterned metallization that is electrically connected to a gate pad of the first one or more of the first power transistor dies, and wherein the metallic island of the patterned metallization that is electrically connected to the source/emitter pad of the first one or more of the first power transistor dies and of the first one or more of the second power transistor dies is interposed between a metallic island of the patterned metallization that is electrically connected to a sense pad of a single one of the first one or more of the second power transistor dies and a metallic island of the patterned metallization that is electrically connected to a gate pad of the first one or more of the second power transistor dies.

14. The power module of claim 13, wherein a single sensor pin is attached to the metallic island of the patterned metallization that is electrically connected to the sense pad of the single one of the first one or more of the first power transistor dies, and wherein a single sensor pin is attached to the metallic island of the patterned metallization that is electrically connected to the sense pad of the single one of the first one or more of the second power transistor dies.

15. The power module of claim 1, wherein a metallic island of the patterned metallization that is electrically connected to a source/emitter pad of the second one or more of the first power transistor dies and of the second one or more of the second power transistor dies is interposed between a metallic island of the patterned metallization that is electrically connected to a sense pad of a single one of the second one or more of the first power transistor dies and a metallic island of the patterned metallization that is electrically connected to a gate pad of the second one or more of the first power transistor dies, and wherein the metallic island of the patterned metallization that is electrically connected to the source/emitter pad of the second one or more of the first power transistor dies and of the second one or more of the second power transistor dies is interposed between a metallic island of the patterned metallization that is electrically connected to a sense pad of a single one of the second one or more of the second power transistor dies and a metallic island of the patterned metallization that is electrically connected to a gate pad of the second one or more of the second power transistor dies.
16. The power module of claim 15, wherein a single sensor pin is attached to the metallic island of the patterned metallization that is electrically connected to the sense pad of the single one of the second one or more of the first power transistor dies, and wherein a single sensor pin is attached to the metallic island of the patterned metallization that is electrically connected to the sense pad of the single one of the second one or more of the second power transistor dies.
17. The power module of claim 1, wherein the first power transistor dies are SiC MOSFET dies, wherein the second power transistor dies are Si IGBT dies, and wherein a diode die is electrically connected antiparallel with each Si IGBT die.
18. The power module of claim 1, wherein the first power transistor dies are SiC MOSFET dies, and wherein the second power transistor dies are Si reverse-conducting IGBT dies.
19. The power module of claim 1, wherein the first power transistor dies are Si MOSFET dies, and wherein the second power transistor dies are Si IGBT dies.
20. The power module of claim 1, wherein the first power transistor dies are Si IGBT dies, and wherein the second power transistor dies are Si reverse-conducting IGBT dies.
21. The power module of claim 1, wherein the first one or more of the first power transistor dies comprises at least three of the first power transistor dies arranged in a row on a same metallic island of the patterned metallization, and wherein a part of the metallic island to which each intermediate one of the at least three of the first power transistor dies is attached is wider than each part of the metallic island to which end ones of the at least three of the first power transistor dies are attached.
22. The power module of claim 1, wherein a first one or more electrical conductors connects a source/emitter pad of each intermediate one of the at least three of the first power transistor dies to an additional metallic island of the patterned metallization, wherein a second one or more electrical conductors connects a source/emitter pad of each end one of the at least three of the first power transistor dies to the additional metallic island, and wherein the first one or more electrical conductors are longer than the second one or more electrical conductors.
23. The power module of claim 1, wherein the second one or more of the first power transistor dies comprises at least three of the first power transistor dies arranged in a row on a same metallic island of the patterned metallization, and wherein a part of the metallic island to which each intermediate one of the at least three of the first power transistor dies is attached is wider than each part of the metallic island to which end ones of the at least three of the first power transistor dies are attached.
24. The power module of claim 1, wherein a first one or more electrical conductors connects a source/emitter pad of each intermediate one of the at least three of the first power transistor dies to an additional metallic island of the patterned metallization, wherein a second one or more electrical conductors connects a source/emitter pad of each end one of the at least three of the first power transistor dies to the additional metallic island, and wherein the first one or more electrical

conductors are longer than the second one or more electrical conductors.

25. The power module of claim 1, wherein a drain/collector pad of the first one or more of the first power transistor dies and a drain/collector pad of the first one or more of the second power transistor dies are attached to a first metallic island of the patterned metallization, wherein a source/emitter pad of the first one or more of the first power transistor dies and a source/emitter pad of the first one or more of the second power transistor dies are electrically connected to a second metallic island of the patterned metallization, wherein a third metallic island of the patterned metallization is electrically connected to the source/emitter pad of the first one or more of the first power transistor dies and of the first one or more of the second power transistor dies, wherein a load current path of the power module includes the first and second metallic islands but excludes the third metallic island.

26. A multi-phase power electronics assembly comprising a plurality of power modules, where each of the power modules supports a different phase and comprises: a substrate comprising a patterned metallization on an electrically insulative body; a plurality of first power transistor dies of a first transistor type attached to the substrate; and a plurality of second power transistor dies of a second transistor type different than the first transistor type attached to the substrate, wherein a first one or more of the first power transistor dies and a first one or more of the second power transistor dies are electrically connected in parallel via the patterned metallization to form a first topological switch, wherein a second one or more of the first power transistor dies and a second one or more of the second power transistor dies are electrically connected in parallel via the patterned metallization to form a second topological switch, wherein the first topological switch and the second topological switch are electrically connected in series via the patterned metallization and enable a load current path for the phase supported by the power module.

27. A power module, comprising: a substrate comprising a patterned metallization on an electrically insulative body; a first power transistor die having a drain/collector pad attached to a first metallic island of the patterned metallization; and a second power transistor die having a drain/collector pad attached to the first metallic island of the patterned metallization, wherein a second metallic island of the patterned metallization is electrically connected to a source/emitter pad at a side of both the first power transistor die and the second power transistor die that faces away from the substrate, wherein a third metallic island of the patterned metallization is electrically connected to the source/emitter pad of the first power transistor die and of the second power transistor die, wherein a load current path of the power module includes the first and second metallic islands but excludes the third metallic island.